Recap: Lecture 22

In Lecture 22, we

- Used separation of variable to find the heat distribution along a rod to be

$$u(x,t) = -\frac{4L}{\pi^2}\sum_{m=1}^{\infty}\frac{(-1)^m}{(2m-1)^2}\mathrm{e}^{-\alpha k_{2m-1}^2 t}\sin(k_{2m-1}x) \quad \text{where} \quad k_n = \frac{n\pi}{L}$$

- Saw that it smooths an initial kink
- This is a *formal solution*. We didn't discuss whether it is a *genuine solution*.

Lecture 23

In Lecture 23, we will:

1. Apply separation of variables to another problem involving the 1D heat equation
2. Start a third problem that involves inhomogeneous BCs

Heat equation in 1D: Problem 2

Consider Problem 1 but where the rod is entirely insulated, so that heat cannot escape

Governing PDE: 1D heat equation for temperature distribution $u(x,t)$ is

$$\frac{\partial u}{\partial t} = \alpha \frac{\partial^2 u}{\partial x^2} \quad \text{for} \quad 0 < x < L$$

Initial condition:

$$u(x,0) = f(x) \quad \text{for} \quad 0 < x < L$$

Boundary conditions: Neumann BCs

$$u_x(0,t) = u_x(L,t) = 0 \quad \text{for} \quad t > 0$$

1. Separate variables to turn single PDE into multiple ODEs *which are tied together by a constant of separation* λ

This is identical to Problem 1:

$$u(x,t) = X(x)\,T(t)$$

$$\Rightarrow \quad X\,\dot{T} = \alpha\,X''\,T$$

$$\Rightarrow \quad \frac{\dot{T}}{\alpha\,T} = \frac{X''}{X} = \lambda$$

$$\Rightarrow \quad \dot{T} - \alpha\lambda\,T = 0 \quad \text{and} \quad X'' - \lambda\,X = 0$$

2. Apply the boundary conditions to find the eigenvalues and eigenfunctions. *This involves a systematic analysis of three possible cases for the constant of separation.*

The BCs gives $X'(0) = X'(L) = 0$.

Case $\lambda = k^2 > 0$:

$$X'' - k^2 X = 0 \quad \Rightarrow \quad X(x) = A\,\mathrm{e}^{k\,x} + B\,\mathrm{e}^{-k\,x}$$

and the BCs again enforce the trivial solution ($A = B = 0$).

2. Apply the boundary conditions to find the eigenvalues and eigenfunctions. *This involves a systematic analysis of three possible cases for the constant of separation.*

The BCs gives $X'(0) = X'(L) = 0$.

Case $\lambda = 0$:

$$X'' = 0 \quad \Rightarrow \quad X(x) = A\,x + B$$

The BCs enforce $A = 0$ but leave B free.

Thus, we choose the eigenfunction $B_0 = 1$ and eigenvalue $\lambda_0 = 0$.

2. Apply the boundary conditions to find the eigenvalues and eigenfunctions. *This involves a systematic analysis of three possible cases for the constant of separation.*

The BCs gives $X'(0) = X'(L) = 0$.

Case $\lambda = -k^2 < 0$:

$$X'' + k^2 X = 0 \quad \Rightarrow \quad X(x) = A\cos(kx) + B\sin(kx)$$

This is similar to Problem 1, but in this case we find the eigenfunctions and eigenvalues

$$X_n(x) = \cos(k_n x), \quad \lambda_n = -k_n^2 \quad (n = 1, 2, \ldots)$$

where $k_n = n\pi / L$.

3. Solve the remaining ODEs for each eigenvalue

$$\boxed{T_n' - \alpha\,\lambda_n\,T_n = 0}$$

This is the same ODE as in Problem 1, so that the solution is

$$T_n(t) = \mathrm{e}^{-\alpha k_n^2 t}$$

We note that this encompasses the $n = 0$ ($\lambda_0 = 0$) case

$$\dot{T}_0(t) = -\alpha\lambda_0 T(t) = 0 \qquad \Rightarrow \qquad T_0(t) = 1$$

4. Apply Superposition Principle to get a series general solution. *The series will have unknown coefficients to be determined.*

We sum all our solutions to get

$$u(x,t) = \frac{1}{2}a_0 + \sum_{n=1}^{\infty} a_n \mathrm{e}^{-\alpha k_n^2 t} \cos k_n x$$

5. Apply the initial conditions to find the series coefficients.

$$u(x,0) = f(x) \quad \Rightarrow$$

$$f(x) = u(x,0) = \frac{1}{2}a_0 + \sum_{n=1}^{\infty} a_n \cos(k_n x)$$

We identify this as the Fourier series of the even periodic extension of $f(x)$ and hence

$$a_0 = \frac{2}{L}\int_0^L f(x)\cos(k_n x)\,dx$$

Let the IC be

$$u(x,0) = f(x) = 1 - \cos\left(\frac{\pi x}{L}\right)$$

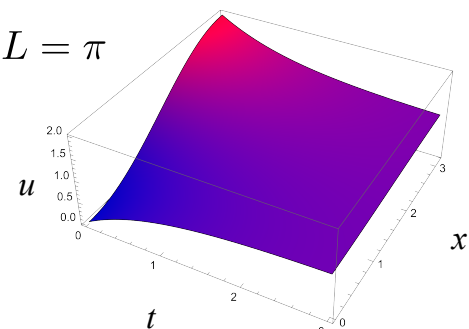
This time we have a Fourier cos series (this is why we put the $1/2$ factor in front of the constant term). We use matching of coefficients (from orthogonality/linear independence) to see

$$a_0 = 2, \qquad a_1 = -1 \qquad \text{and} \qquad a_n = 0 \quad \text{for} \quad n \neq 1$$

The final solution is

$$u(x,t) = 1 - \mathrm{e}^{-\alpha k_1^2 t} \cos(k_1 x)$$

$$\alpha = 1, \quad L = \pi$$



Heat equation in 1D: Problem 3

Consider a rod which is initially at zero temperature. The temperature at one end is heated instantaneously and held at a higher constant temperature equal to the rod length, while the other is held at the initial temperature.

Governing PDE: 1D heat equation for temperature distribution $u(x,t)$ is

$$\frac{\partial u}{\partial t} = \alpha \frac{\partial^2 u}{\partial x^2} \quad \text{for} \quad 0 < x < L$$

Initial condition:

$$u(x,0) = 0 \quad \text{for} \quad 0 < x < L$$

Boundary conditions: Dirichlet BCs

$$u(0,t) = 0, \quad u(L,t) = L \quad \text{for} \quad t > 0$$

Inhomogeneous BCs

- Express the solution as
$$u(x,t) = u_S(x) + u_T(x,t)$$
- u_S is the *steady-state solution*, which is independent of time
- u_T is the *transient solution*, such that, as $t \to \infty$,
$$u_T(x,t) \to 0 \quad \Rightarrow \quad u(x,t) \to u_S(x)$$

Steady state solution, $u_S(x)$

Obtain governing ODE

Substitute $u_S(x)$ into PDE to get

$$\frac{\partial}{\partial t}u_S(x) = \frac{\partial^2}{\partial x^2}u_S(x) \qquad \Rightarrow \qquad u_S'' = 0$$

Impose inhomogeneous BCs on $u_S(x)$

$$u_S(0) = 0 \qquad \text{and} \qquad u_S(L) = L$$

Solution is

$$u_S(x) = x$$

Transient solution $u_T(x,t)$: BCs and ICs

BCs are homogeneous for $u_T(x,t)$

$$u(0,t) = u_T(0,t) + u_S(0) = 0 \qquad \Rightarrow \qquad u_T(0,t) = 0$$

$$u(L,t) = u_T(L,t) + u_S(L) = L \qquad \Rightarrow \qquad u_T(L,t) = 0$$

IC is a function of both $u(x,0)$ and $u_S(x)$

$$u(x,0) = f(x) = u_S(x) + u_T(x,0) \qquad \Rightarrow \qquad u_T(x,0) = g(x)$$

$$\text{where} \quad g(x) = f(x) - u_S(x) = -x$$

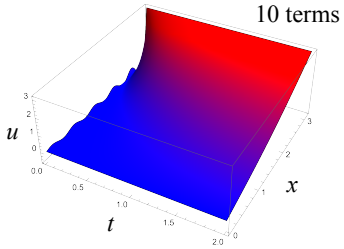
Problem 3: Solution

The full solution is

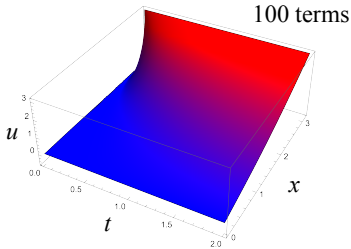
$$u(x,t) = u_S(x) + u_T(x,t) = x + \frac{2L}{\pi}\sum_{n=1}^{\infty}\frac{(-1)^n}{n}\mathrm{e}^{-\alpha\, k_n^2\, t}\sin\left(k_n x\right)$$

$\alpha = 1, \quad L = \pi$

10 terms



100 terms



Zoom in on initial condition 100 terms

